METRIC

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DETAIL SPECIFICATION

TRANSMISSION, HYDRAULIC, MODEL XTG-250-1A

This specification is approved for use within U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of hydraulic transmission and major components thereof, for use in tracked military vehicles.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2520

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SPECIFICATIONS

FEDERAL

A-A-50271	- Plates; Identification.

DEPARTMENT OF DEFENSE

MIL-PRF-2104	- Lubricating Oil, Internal Combustion Engine, Heavy-Duty.
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.

(Unless otherwise indicated, copies of the above specifications and standards are available from the Defense Printing Service Detachment Office (Customer Service), 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

8356100	- Main Assembly - Power Train XTG-250-1A.
8356299	- Kit - Oil Filler.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

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3.2 Materials, design, and manufacturing processes. Unless otherwise specified, the materials, design, and manufacturing process selection is the prerogative of the contractor as long as all articles submitted to the Government fully meet the operating, interface, support and ownership, and operating environment requirements specified.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Construction. Transmission construction and assembly shall be in accordance with Drawing 8356100. Unless otherwise specified (see 6.2), oil filler, Drawing 8356299, shall not be furnished with each transmission (see 4.5.3).

3.4 Operating requirements.

3.4.1 Functional operation. The transmission shall operate in all shift and steer positions without indication of mechanical malfunction. With the transmission not running, the torque effort required to move the selector shaft to any shift position shall not exceed 10.2 Newton meters (N·m) (90 inch-pounds (in-lb)) and the torque effort required to move the pivot shaft to any shift position shall be not less than 3.4 N·m (30 in-lb) (see 4.5.5.1).

3.4.1.1 Brake adjustment. The transmission brakes shall be adjusted such that, when the brake is applied with 136 N·m (100 foot pounds (ft-lb)) torque, the position indicator shall be in alignment with the “apply” mark located on the top cover; and when released the indicator shall be in alignment with the “off” mark (see 4.5.5.2).

3.4.2 Fourth range stall. With range selector in fourth range, throttle valve (TV) full on, output shafts stalled, and input torque gradually increased to between 494 N·m (364 ft-lb) and 499 N·m (368 ft-lb); transmission performance shall be in accordance with table I (see 4.5.5.3).

TABLE I. Fourth range stall performance.

Item	Minimum	Maximum
Input speed revolutions per minute (rpm)	1750	1900
Output torque N·m (ft-lb)	3254 (2400)	
Oil pressure kPa (psi) 1/		
Lube	138 (20)	276 (40)
Main	1310 (190)	1482 (215)
Right Hand (RH) output clutch	Actual main minus 172 (25)	
Left Hand (LH) output clutch	Actual main minus 172 (25)	

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TABLE I. Fourth range stall performance - Continued.

Item	Minimum	Maximum
High range clutch	Actual main minus 90 (13)	
Cooler flow L/min (gal/min) 2/	62.5 (16.5)	
Oil temperature cooler °C (°F) 3/	93 (200)	107 (225)

1/ kPa (psi) = kilopascals (pounds per square inch)
2/ L/min (gal/min) = liters per minute (gallons per minute)
3/ °C (°F) = degrees Celsius (degrees Fahrenheit)

3.4.3 Fourth range steer. With range selector in fourth range and an input speed of 600-650 rpm, right and left output shafts shall rotate as specified in table II (see 4.5.5.3).

TABLE II. Fourth range steer.

Pivot valve position	Steer	Direction of RH & LH output shaft rotation	Speeds of RH & LH output shafts
Pivot	Full right	Same	Different
Pivot	Full left	Same	Different

3.4.4 Lockup clutch operation.

3.4.4.1 Lockup clutch engagement - no load. With range selector in third range, (TV) full “on”, and no load on output shaft; the transmission shall automatically engage the lockup clutch at an output speed of not less than 250 rpm and not more than 295 rpm. At the lockup clutch engagement point, the output pump shall have a minimum pressure of 1310 kPa (190 psi) (see 4.5.5.4).

3.4.4.2 Lockup clutch load. With range selector in third range, TV “on”, an input speed of 2395-2405 rpm, and an output load of not less than 3390 N·m (2500 ft-lb) and not more than 3796 N·m (2800 ft-lb); the lockup clutch pressure shall be not less than 1034 kPa (150 psi), and not more than 1241 kPa (180 psi), and output speed shall be not less than 323 rpm, and not more than 343 rpm (see 4.5.5.4).

3.4.4.3 Lockup clutch disengagement - no load. With range selector in third range, no TV valve pressure, lockup clutch engaged, and no load on output shafts; the transmission shall automatically disengage the lockup clutch at an output speed of not less than 105 rpm, and not more than 160 rpm (see 4.5.5.4).

3.4.5 Second range land steer. When operating from neutral steer position, to full steer, and back to neutral; oil pressures shall be as specified in table III. When in full steer, both outputs shall rotate in the same direction, but at different speeds (see 4.5.5.5).

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TABLE III. Second range land steer - input speed 1495 to 1505 rpm.

Pressure kPa (psi)	Neutral steer	Full steer	Neutral steer
RH steer			
RH output clutch	--	--	--
RH steer clutch	--	1034 (150) min	--
RH coolant	0-103 (0-15)	28-207 (4-30) above Neutral pressure	0-103 (0-15)
LH steer			
LH output clutch	--	--	--
LH steer clutch	--	1034 (150) min	--
LH coolant	0-103 (0-15)	28-207 (4-30) above Neutral pressure	0-103 (0-15)

3.4.6 Second range pivot steer. When operating from neutral steer position, to 8 degrees (°) from neutral position, to full steer, and back to neutral; oil pressures shall be as specified in table IV. When in full steer, output shafts shall rotate in opposite directions, but at different speeds (see 4.5.5.5).

TABLE IV. Second range pivot steer - input speed 1495 to 1505 rpm.

Pressure kPa (psi)	Neutral steer	8° steer	Full steer	Neutral steer
RH steer				
RH output clutch	--	0	--	--
RH reverse clutch	--	103-241 (15-35)	1034 (150) min	--
RH coolant	0-103 (0-15)	--	207 (30) max above Neutral pressure	0-103 (0-15)
LH steer				
LH output clutch	--	0	--	--
LH reverse clutch	--	103-241 (15-35)	1034 (150) min	
LH coolant	0-103 (0-15)	--	207 (30) above max Neutral pressure	0-103 (0-15)

3.4.7 Service brake capacity. With the selector range in first range, TV full “on”, and RH output shaft coupled to and stalled by the dynamometer, the input speed shall be increased until RH output shaft torque is between 4732 and 4760 N·m (3490 and 3510 ft-lb). Then with 129 to 142 N·m (95 to 105 ft-lb) torque applied to RH brake shaft and dynamometer released, the RH brake shall hold transmission output without movement. Under similar conditions, the LH brake shall hold transmission output without movement (see 4.5.5.6).

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3.4.7.1 Brake coolant pump pressure. With transmission output shafts uncoupled from the dynamometer, range selector in fourth range, and input speed of 900 to 1000 rpm, and the "pivot" valve in the "land" position; the RH and LH coolant pressure shall not exceed 172 kPa (25 psi). With the RH brake partially applied, the right hand coolant pump pressure shall rise to 345 kPa (50 psi) min, to 689 kPa (100 psi) max, and return to 172 kPa (25 psi) max when the brake is released. With the LH brake partially applied, the LH coolant pump pressure shall rise to 345 kPa (50 psi) min, to 689 kPa (100 psi) max, and return to 172 kPa (25 psi) max when the brake is released (see 4.5.5.7).

3.4.8 Reverse range stall. With the range selector in reverse-one (R1) range, output shafts stalled by the dynamometer, full TV, and the input torque gradually increased to 157-163 N·m (116-120 ft-lb); the transmission performance shall be as specified in table V (see 4.5.5.8).

TABLE V. Reverse range stall.

Item	Output torque N·m (ft-lb)	Oil pressure kPa (psi)
Main	5966 (4400) minimum	965 (140) min -1206 (175) max
RH reverse clutch	5966 (4400) minimum	Actual main -69 (10) min
LH reverse clutch	5966 (4400) minimum	Actual main -69 (10) min

3.4.9 Fourth range friction loss. With range selector in fourth range, an input speed of 1995-2005 rpm, no TV, output shafts uncoupled, and service brakes released; there shall be not more than 136 N·m (100 ft-lb) input torque (see 4.5.5.9). The cooler "in" oil temperature shall be between 66 and 107°C (150 and 225°F).

3.4.10 Downshift inhibitor. A fourth to third range downshift shall be permitted by the downshift inhibitor when input speed is reduced slowly, with an initial input speed between 1950 and 2050 rpm in fourth range, no TV, and output shafts uncoupled. A first range to neutral shift shall be permitted by the downshift inhibitor under the same conditions. Downshift input speed shall be within the speed values specified in table VI. The lockup clutch shall be applied prior to the downshift. Lockup pressure shall drop momentarily as the fourth to third range shift is made and return to its original value when the shift is completed (see 4.5.5.10).

TABLE VI. Shift values.

	Input speed (rpm)	
Downshift range	min	max
4th to 3rd	1200	1520
1st to neutral	1240	1500

3.4.11 Speedometer drive. The speedometer drive gear shall rotate when the transmission output shaft is turning (see 4.5.3 and 4.5.5.11).

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3.4.12 Low speed oil pressure and temperature. With an input speed of 600-650 rpm, no load on the output, and the "pivot" valve in the "land" position, main oil pressure and cooler "in" oil temperature shall be as specified in table VII (see 4.5.5.12).

TABLE VII. Low speed oil pressure and temperature.

Range	Main oil pressure	Cooler "in" oil temp
4th range	648 kPa (94 psi) min	88 - 107°C (190 - 225°F)
3rd range	896 kPa (130 psi) min	77 - 107°C (170 - 225°F)
2nd range	979 kPa (142 psi) min	77 - 107°C (170 - 225°F)
1st range	1172 kPa (170 psi) min	77 - 107°C (170 - 225°F)
Reverse 2 range	1034 kPa (150 psi) min	77 - 107°C (170 - 225°F)
Reverse 1 range	862 kPa (125 psi) min	77 - 107°C (170 - 225°F)

3.5 Interface requirements. The assembly shall conform to the interface envelope dimensions, mounting and electrical interfaces in the applicable drawing (see 4.6).

3.6 Support and ownership requirements. The assembly shall possess the following life cycle ownership characteristics (see 4.7).

3.6.1 Safety. The transmission shall pose no hazards, physically or electrically to personnel (see 4.7.1).

3.6.2 Reliability. After proper installation in the vehicle for which the transmission has been procured, the transmission shall require no repair and no maintenance other than filter changes, oil changes, oil additions, and brake adjustments during vehicle operation at a Government proving ground for distances up to 8000 kilometers (km) (5000 miles) (see 4.7.2).

3.6.3 Leakage. After dynamometer operation, transmission shall evidence no leakage (see 4.5.3, 4.7.3, and 6.6.1).

3.6.4 Exterior surface treatment. All exposed surfaces of the transmission and its components shall be cleaned, painted or treated for corrosion resistance as specified on applicable drawings (see 4.5.3).

3.6.5 Nameplate. Unless otherwise specified on the applicable drawing, the nameplate, data and instruction plates shall conform to the applicable provisions of A-A-50271 (see 4.5.3).

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4. VERIFICATION

4.1 Classification of inspections. Inspections specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 First article inspection. When specified (see 6.2), first article inspection shall consist of one complete transmission assembly when a first article is required (see 3.1). This inspection shall include the examination of 4.4 and the tests of 4.5.5.1 thru 4.7.4.

4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.4 (see table VIII) and the tests of 4.5.5.1 thru 4.7.4 (see table IX).

4.4 Examination. Each transmission shall be examined for compliance with the requirements specified in 3.3, 3.4.1, 3.4.11, and 3.6.3 thru 3.6.5 (see table VIII). Noncompliance with any of the specific requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

TABLE VIII. Classification of defects.

Category	Defect	Method of inspection
Major:		
101	Dimensions affecting interchangeability, not within tolerance (see 3.3).	Gage or Scale
102	Malfunction in control levers or valves (see 3.4.1).	Functional/ Visual
103	Malfunction in speedometer drive (see 3.4.11).	Visual
104	Oil leakage (see 3.6.3).	Visual
Minor:		
201	Dimensions not affecting interchangeability, not within tolerance (see 3.3).	Gage or Scale
202	Improper application or color of paint (see 3.6.4).	Visual
203	Improper or illegible marking (see 3.6.5).	Visual

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TABLE IX. Classification of inspection.

Title	Requirement	Veri-fication	First article	Conformance
Materials, design and construction	3.2		X	X
Operating requirements:	3.4	4.5.5	X	X
Functional	3.4.1	4.5.5.1	X	X
Brake adjustment	3.4.1.1	4.5.5.2	X	X
Fourth range stall	3.4.2	4.5.5.3	X	X
Fourth range steer	3.4.3	4.5.5.3	X	X
Lockup clutch engagement-no load	3.4.4.1	4.5.5.4	X	X
Lockup clutch load	3.4.4.2	4.5.5.4	X	X
Lockup clutch disengagement-no load	3.4.4.3	4.5.5.4	X	X
Second range land steer	3.4.5	4.5.5.5	X	X
Second range pivot steer	3.4.6	4.5.5.5	X	X
Service brake capacity	3.4.7	4.5.5.6	X	X
Brake coolant pump pressure	3.4.7.1	4.5.5.7	X	X
Reverse range stall	3.4.8	4.5.5.8	X	X
Fourth range friction loss	3.4.9	4.5.5.9	X	X
Down inhibitor	3.4.10	4.5.5.10	X	X
Speedometer drive	3.4.11	4.5.5.11	X	X
Low speed oil pressure and temperature	3.4.12	4.5.5.12	X	X
Interface requirements:	3.5	4.6		
Support and ownership requirements:	3.6	4.7		
Reliability	3.6.2	4.7.2	X	X
Leakage	3.6.3	4.7.3	X	X
Exterior surface	3.6.4	4.5.3	X	X
Nameplate	3.6.5	4.7.4	X	X

4.5 Method of inspection.

4.5.1 Verification methods. The types of verification methods included in this section are visual inspection, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously approved or previously qualified designs.

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4.5.2 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipments including the application of statistical process control, tool control, or cost-effective sampling procedures, to verify performance. See the contract for alternatives that replace verifications required by this specification.

4.5.3 Defects. Conformance to 3.3, 3.4.1, 3.4.11, and 3.6 thru 3.6.5 shall be determined by examination for the defects listed in table VIII. Examination shall be visual, tactile, or by measurement with SIE.

4.5.4 Oil. Oil conforming to grade 1 of MIL-L-21260 shall be used during the transmission dynamometer operation and production surveillance runs. Unless otherwise specified, the oil temperature shall be maintained between 71 and 121°C (160 and 250°F) during dynamometer operation. During the dynamometer operation oil temperature shall be not more than 149°C (300°F). The transmission, as installed in vehicle, shall be filled to operating level with oil conforming to grade 10 of MIL-PRF-2104 when operating in ambient air temperature above −18°C (0°F). Oil conforming to MIL-L-46167 shall be used when operating in ambient air temperature below −18°C (0°F). Oil level shall be maintained within the operating range on the oil level indicator. After the transmission is filled within the operating range, operate the transmission as required to bring oil to operating temperature, then stop for 3 minutes before checking for proper oil level.

4.5.5 Operating requirements verification.

4.5.5.1 Functional test. The range selector lever shall be placed in the third range position. With the output shafts stalled by the dynamometer, increase input speed to 980-1020 rpm and run at that speed for a minimum of one minute. Run at 600-650 rpm input speed and operate at all steer and shaft positions for any indication of mechanical malfunction.

CAUTION: Transmission shall not be shifted into “land” or “pivot” positions until the output dynamometer is released and the input speed is reduced to 350 rpm maximum.

With transmission not running, measure torque required to move selector shaft and pivot shaft, as specified, to determine conformance to 3.4.1. Any mechanical malfunction or torque values not in accordance with 3.4.1 shall constitute failure.

4.5.5.2 Brake adjustment test. Brakes shall be applied with 136 N·m (100 ft-lb) of torque and alignment of the position indicator shall be verified, to determine conformance to 3.4.1.1 (see 6.5).

4.5.5.3 Fourth range stall and steer test. The range selector shall be placed in fourth range and the throttle valve full “on”. With output shafts stalled, input torque shall be gradually

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increased to between 494 and 499 N·m (364 and 368 ft-lb) and transmission performance compared to table I to determine conformance to 3.4.2. Subsequently, the output shafts shall be released, input speed reduced to 600-650 rpm and output shafts observed. Performance shall be compared to table II and verified to determine conformance to 3.4.3.

4.5.5.4 Lockup clutch operation test. The range selector shall be placed in third range, and the throttle valve full "on". With no load on output shaft, lockup clutch engagement speed and output pump pressure shall be observed to determine conformance to 3.4.4.1. With the range selector still in third range and the throttle valve full "on" the input speed shall be adjusted to 2395-2405 rpm and output load to 3390-3796 N·m (2500-2800 ft-lb). Lockup clutch pressure and output speed shall be observed to determine conformance to 3.4.4.2. With the range selector still in third range, no throttle valve pressure, lockup clutch engaged and no load on output shafts, the output speed at lockup clutch automatic disengagement shall be observed and verified to determine conformance to 3.4.4.3.

4.5.5.5 Second range land and pivot steer test. Transmission shall be operated from neutral steer position, to full steer, and back to neutral. Performance shall be compared to table III to determine conformance to 3.4.5. Subsequently, transmission shall be operated from neutral steer position, to 8° from neutral, to full steer, and back to neutral. Performance shall be compared to table IV and verified to determine conformance to 3.4.6.

4.5.5.6 Service brake capacity test. The range selector shall be placed in first range; TV full "on"; and RH output shaft coupled to, and stalled by, the dynamometer. With input speed increased until RH output shaft torque is between 4732-4760 N·m (3490-3510 ft-lb), 129-142 N·m (95-105 ft-lb) torque shall be applied to RH brake shaft. As dynamometer is released, the RH brake shall hold transmission output without movement. To determine conformance to 3.4.7, test shall be repeated on LH brake. Brakes that fail to hold the transmission output without movement shall be considered a failure.

4.5.5.7 Brake coolant pump pressure test. The range selector shall be placed in fourth range and transmission output shafts uncoupled from the dynamometer. With an input speed of 900 to 1000 rpm and the "pivot" valve in the "land" position, the RH and LH coolant pressure shall not exceed 172 kPa (25 psi). The RH and LH brakes shall be alternately applied and released, and the coolant pump pressure rise and return observed to determine conformance to 3.4.7.1.

4.5.5.8 Reverse range stall test. The range selector shall be placed in reverse-one (R1) range and the throttle valve full "on". With output shafts stalled by the dynamometer and input torque gradually increased to 157-163 N·m (116-120 ft-lb), transmission performance shall be compared to table V and verified to determine conformance to 3.4.8.

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4.5.5.9 Fourth range friction loss test. The range selector shall be placed in fourth range, no throttle valve, and output shafts uncoupled. With input speed of 1995-2005 rpm and service brakes released, the input torque and cooler “in” oil temperature shall be observed and verified to determine conformance to 3.4.9.

4.5.5.10 Downshift inhibitor test. With input speed between 1950 and 2050 rpm in fourth range, no throttle valve, and output shafts uncoupled, input speed shall be slowly reduced until downshift occurs. The operation shall be repeated in first range. Shift values and lockup pressure shall be observed and verified to determine conformance to 3.4.10.

4.5.5.11 Speedometer drive test. During any appropriate test, speedometer drive gear shall be observed to determine conformance to 3.4.11.

4.5.5.12 Low speed oil pressure and temperature test. With input speed of 600-650 rpm, no load on output, and “pivot” valve in the “land” position, main oil pressure and cooler “in” oil temperature shall be observed and verified to determine conformance to 3.4.12.

4.6 Interface requirements verification. Verify the envelope dimensions and mounting interface by measurements and placement.

4.7 Support and ownership requirements verification.

4.7.1 Safety inspection. To determine conformance to 3.6.1, assemblies shall be inspected for hazardous burrs, sharp edges, foreign materials, or other imperfections that pose physical danger to an installer/operator. Inspect the assembly for exposed, frayed, unsecured, or otherwise improperly protected wires that pose electrical danger to the installer/ operator. Any defect shall result in rejection of the transmission.

4.7.2 Reliability. To determine conformance to 3.6.2, vehicle operation shall include any of the types of operation, terrain conditions, and ambient temperature to which a military vehicle of any kind is subjected, in accordance with the provisions of the specification covering the vehicle.

4.7.3 Leakage. Subsequent to dynamometer testing, transmission shall be examined to determine conformance to 3.6.3. Leakage shall constitute failure.

4.7.4 Identification. To determine conformance to 3.6.5, assemblies shall verify the presence of the required markings on the assembly. After performing all operating requirements tests in 4.5.5.1 thru 4.5.5.12, re-inspect the assembly markings for readability.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The transmission covered by this specification is intended for production use, as spares, or as replacements in military combat and tactical vehicles. The transmission is military unique because of the extreme environmental conditions encountered. Commercial components are not designed to withstand such extreme environmental conditions and would experience failure.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).
c. If a first article sample is required (see 3.1).
d. Oil filler, if required (see 3.3).
e. When first article inspection is required (see 4.2).
f. Packaging requirements (see 5.1).

6.3 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

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6.4 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, Government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from table IX and their designated frequency based on a risk assessment for the procurement.

6.5 Brake adjustment. Before or during dynamometer operation, 136 N·m (100 ft-lb) torque should be applied to the brake shaft. To adjust brake, remove brake inspection cover and turn adjusting screw until proper indicator alignment is obtained. Each brake should be adjusted separately (see 4.5.5.2).

6.6 Definitions.

6.6.1 Leakage. The following definitions are established as guidelines governing leakage:

a. Weep - Any evidence of fluid beyond a seal or splitline.
b. Seep - Any evidence of fluid beyond a seal or splitline that does not result in the formation of a droplet.
c. Leak - Any evidence of fluid beyond a seal or splitline that results in the formation of a droplet.
d. Drip - Any evidence of fluid beyond a seal or splitline where droplets form and fall.

6.6.2 Transmission orientation. The terms LH and RH derive from the following orientation:

a. The “front” of the transmission is the end which mounts next to the engine.
b. The “rear” of the transmission is the end opposite from the front.
c. The “right” side is the side on the viewer’s right as he faces the transmission from the rear.
d. The “left” side is the side on the viewer’s left as he faces the transmission from the rear.
e. These definitions are established relative to the transmission only and not relative to any vehicle installation definitions.

6.7 Recycled materials. The use of recycled materials, which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item, will be encouraged (see 3.2).

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6.8 Subject term (key word) listing.

Brake adjustment
Downshift inhibitor
Fourth range steer
Full steer
Functional operation
Leakage
Lockup clutch operation
Neutral steer

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project 2520-0017)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-62030E(AT)	**2. DOCUMENT DATE *(YYMMDD)*** 980908

3. DOCUMENT TITLE TRANSMISSION, HYDRAULIC, MODEL XTG-250-1A

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	**b. ORGANIZATION**	
c. ADDRESS *(Include Zip Code)*	**d. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** ***(If applicable)***	**7. DATE SUBMITTED *(YYMMDD)***

8. PREPARING ACTIVITY

a. NAME	**b. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** **(810) 574-8745** **786-8745**
c. ADDRESS *(Include Zip Code)* **Commander** **U.S. Army Tank-automotive and Armaments Command** **ATTN: AMSTA-TR-E/BLUE** **Warren, MI 48397-5000**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403** **Falls Church, VA 22041-3466** **Telephone (703) 756-2340 AUTOVON 289-2340**

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